Introduction to Quantum Computing

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Introduction

- First of all, I am not an expert on the Quantum Computing.
- The purpose is
- (1) to introduce basics of Quantum Computing to student
  - It is possible that you already know better than me on the subject.
- (2) to be on the same ground to start a further discussion on the application of Quantum Computing to nuclear physics.

- For this purpose, I will only skim through some basics. I borrowed many materials from existing lectures, presentations, papers.

- One specific goal is to understand a paper about quantum computing on nuclear physics.

Why Quantum computer

- Quantum computing: A type of computing where the programming model f ollows principles of quantum mechanics.
- Why needed? There are problems whose computing requirement grows non-polynomial way.

- For example, classical Ising model

$$E(\{s\}) = -J\sum_{\langle ij\rangle} s_i s_j, \quad s_i \in \{-1,1\}$$

$$Z = \sum_{\{s\}} \exp\left(-\frac{E(\{s\})}{k_B T}\right)$$

$$\langle E\rangle(T) = \frac{1}{Z}\sum_{\{s\}} E(\{s\}) \exp\left(-\frac{E(\{s\})}{k_B T}\right)$$

- All possible spin configuration on N^3 lattice is 2^{N^3}
- Suppose N=10 and n floating point operation per co nfiguration
- ~ 10^{301} n floating point operation
- If 10^{18} Flops/s → 10^{283} n seconds.
- Thus exact calculation is not possible.(MC approac h can be used.)

Why Quantum computer

- Quantum computer can be faster than classical computer in some problems. (Quantum supremacy)
- Short history:
  - P. Benioff (1980), "The Computer as a Physical System: A Microscopic Quantum Mechanical Hamiltonian Model of Computers as Represented by Turing Machines"
  - R.P. Feynman (1982), "Simulating Physics with Computers":
    "Can a *universal* quantum simulator be built?"
  - S. Lloyd (1996), "Universal Quantum Simulators":
    "Yes"

Single Qubit

- Data are represented by qubits in Quantum Computer.

  - Classical computing: bits

    $$b \in \{0,1\}$$

  - Quantum computing: qubits

    $$|q\rangle \in C^2$$

  - Computational basis:

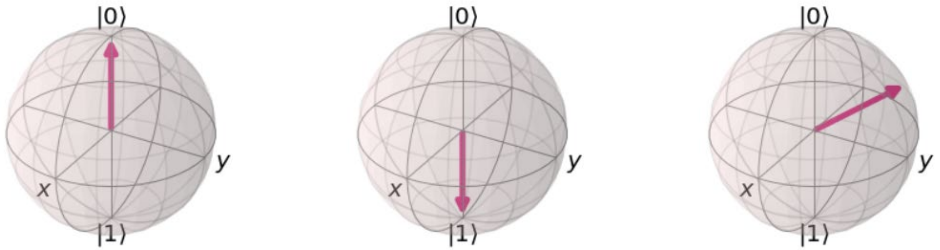
    $$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \qquad |1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

  - An arbitrary state is a *superposition*

    $$|q\rangle = \alpha|0\rangle + \beta|1\rangle \qquad |\alpha|^2 + |\beta|^2 = 1$$

    or

    $$|q\rangle = e^{i\gamma}\left(\cos\frac{\theta}{2}|0\rangle + e^{i\phi}\sin\frac{\theta}{2}|1\rangle\right)$$



$|q\rangle = |0\rangle$ $\qquad$ $|q\rangle = |1\rangle$ $\qquad$ $|q\rangle = \alpha|0\rangle + \beta|1\rangle$

Bloch representation

Difference between classical bit and qubit

- Single Classical bit can represent 0 or 1
- Qubit can represent arbitrary superposition
  → complex numbers (or phase angles)

Multiple Qubits

- N classical bits can repre sent 2^N different states
- 8 bits → 256 integers

- N qubits (tensor product of qubits) can represent (C^2)^N different states
- 8 qubits → 256 complex numbers

$$|q_1\rangle = \alpha_0|0\rangle + \alpha_1|1\rangle = \sum_{i=0,1} \alpha_i|i\rangle$$

$$|q_2\rangle = \alpha_{00}|0\,0\rangle + \alpha_{01}|0\,1\rangle + \alpha_{10}|1\,0\rangle + \alpha_{11}|1\,1\rangle = \sum_{i,j} \alpha_{ij}|ij\rangle\,,$$

$$\begin{array}{l} 00000000 \\ 00000001 \\ \vdots \\ 01101010 \\ 01101011 \\ 01101100 \\ \vdots \\ 11111110 \\ 11111111 \end{array} \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ 1 \\ 0 \\ \vdots \\ 0 \\ 0 \end{bmatrix} . \qquad \begin{array}{l} 00000000 \\ 00000001 \\ \vdots \\ 01101010 \\ 01101011 \\ 01101100 \\ \vdots \\ 11111110 \\ 11111111 \end{array} \begin{bmatrix} c_0 \\ c_1 \\ \vdots \\ c_{106} \\ c_{107} \\ c_{108} \\ \vdots \\ c_{254} \\ c_{255} \end{bmatrix}$$

$$\sum_{i=0}^{255} |c_i|^2 = 1.$$

Superposition of 256 basis states

This requires entanglement between qubits

Measurement collapse to
One classical bits states

DiVincenzo's requirements

8.1 DiVincenzo's Five Requirements for the Quantum Computation

1 A scalable physical system with well-chracterized qubits

2 A qubit-specific measurement capability

3 The ability to initialize the state of qubits a simple fiducial state ($|00\ldots00\rangle$)

4 A "universal" set of quantum gates

5 Long relevant decoherence times, much longer than the gate operation time

6 (For Quantum Communication) The ability to interchange between stationary vs flying qubits.

7 (For Quantum Communication) The ability to transmit flying qubits between specific locations reliably.

Types of quantum computers

- Any quantum system can be a candidate of building block of quantum computers:
  - usually two-level system

- Simulator (Analog quantum computers) :
  - building an artificial many-body system with a similar Hamiltonian
  - Limited and specific set of controls
  - Ultracold atoms, Spin qubits , Superconducting qubits
  - Used as proxies (or simulators) for the many-body system one we wants to unde rstand.

- Computers (Digital (gate-based) quantum computers):
  - individual control over particles/degrees of freedom by gate-based operation.
  - Internally, each gate is performed using an analog description.
  - Universal

Universal quantum gates

- Classical computers can be built from elementary or basic logical gates.
- Any Vector valued Boolean functions can be expressed as a combination of elementa ry gates.

$$f : \{0,1\}^l \to \{0,1\}^m.$$

$$\mathrm{NOT}: \{0,1\} \to \{0,1\}, \qquad b \mapsto \mathrm{NOT}(b) = \bar{b}.$$

$$\mathrm{OR}: \{0,1\}^2 \to \{0,1\}, \qquad b_1 b_2 \mapsto \mathrm{OR}(b_1 b_2)$$

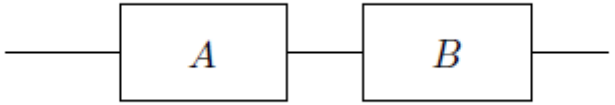
$$\mathrm{AND}: \{0,1\}^2 \to \{0,1\}, \qquad b_1 b_2 \mapsto \mathrm{AND}(b_1 b_2)$$

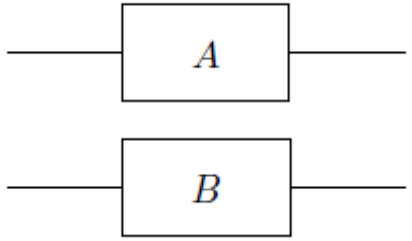
- Quantum computers can be built from universal quantum gates.
- Gate Operation on qubits must be unitary

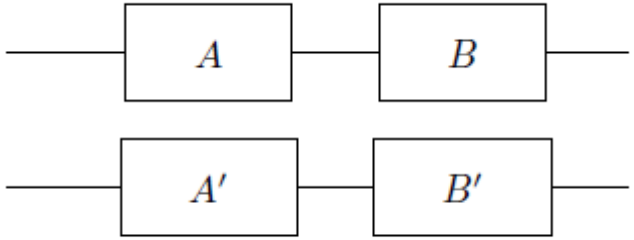
Universal quantum gates

- Solovay-Kitaev theorem:
  - any unitary operation acting on n-qubits can be written as a sequence of universal gate set.
  - Universal gate set : one-qubit rotations Rx, Ry, Rz + two-qubit gate “CNOT”

- Sequential operation on a single qubit = Unitary matrix
- Parallel operations on multiple qubits = tensor product of matrices







$$(B \star A) \otimes (B' \star A') = (B \otimes B') \star (A \otimes A').$$

Single qubit Gates

- Pauli matrices

$$I \equiv \sigma_0 \equiv \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \qquad X \equiv \sigma_1 \equiv \sigma_x \equiv \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

$$Y \equiv \sigma_2 \equiv \sigma_y \equiv \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \qquad Z \equiv \sigma_3 \equiv \sigma_z \equiv \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

- Pauli X acts as a NOT gate: $|0\rangle \rightarrow |1\rangle$ and $|1\rangle \rightarrow |0\rangle$

Phase — S — $\begin{bmatrix} 1 & 0 \\ 0 & i \end{bmatrix}$

$\pi/8$ — T — $\begin{bmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{bmatrix}$

Name	Symbol	Matrix	Name	Symbol	Matrix	Name	Symbol	Matrix
X	X	$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$	Y	Y	$\begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$	Z	Z	$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$
Hadamard	H	$\frac{1}{\sqrt{2}}\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$	Phase	$P(\varphi)$	$\begin{bmatrix} 1 & 0 \\ 0 & e^{i\varphi} \end{bmatrix}$	Universal	$U(\theta, \phi, \lambda)$	$\begin{bmatrix} \cos\left(\frac{\theta}{2}\right) & -e^{i\lambda}\sin\left(\frac{\theta}{2}\right) \\ e^{-i\phi}\sin\left(\frac{\theta}{2}\right) & e^{i(\phi+\lambda)}\cos\left(\frac{\theta}{2}\right) \end{bmatrix}$
X-rotation	$R_x(\theta)$	$\begin{bmatrix} \cos\left(\frac{\theta}{2}\right) & -i\sin\left(\frac{\theta}{2}\right) \\ -i\sin\left(\frac{\theta}{2}\right) & \cos\left(\frac{\theta}{2}\right) \end{bmatrix}$	Y-rotation	$R_y(\theta)$	$\begin{bmatrix} \cos\left(\frac{\theta}{2}\right) & -\sin\left(\frac{\theta}{2}\right) \\ \sin\left(\frac{\theta}{2}\right) & \cos\left(\frac{\theta}{2}\right) \end{bmatrix}$	Z-rotation	$R_z(\theta)$	$\begin{bmatrix} e^{-i\frac{\theta}{2}} & 0 \\ 0 & e^{i\frac{\theta}{2}} \end{bmatrix}$

Single qubit Gates

Hadamard $\quad -\boxed{H}- \quad \frac{1}{\sqrt{2}}\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$

Hadamard gate : pure (basis) state → mixed (Hadamard basis) state → pure state

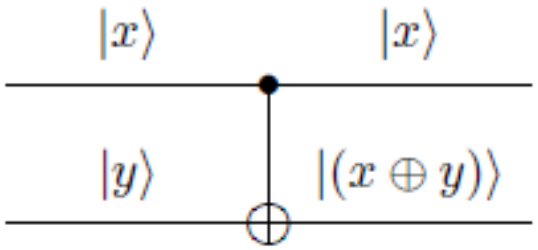
$$H|0\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle\,, \qquad H|1\rangle = \frac{1}{\sqrt{2}}|0\rangle - \frac{1}{\sqrt{2}}|1\rangle\,,$$

$$H|0\rangle = |+\rangle\,, \qquad H|1\rangle = |-\rangle\,.$$

$$H|q\rangle_H = H(\alpha_+|+\rangle + \alpha_-|-\rangle) = \alpha_+ H|+\rangle + \alpha_- H|-\rangle = \alpha_+|0\rangle + \alpha_-|1\rangle$$

Two qubit Gates

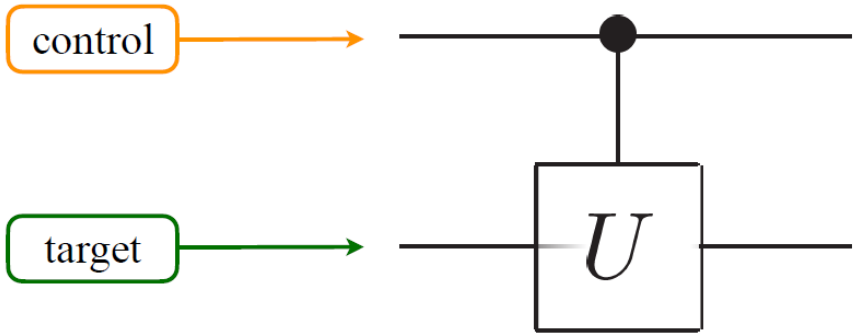
- CNOT gate : control bit



$\oplus$ is the binary exclusive or operation.

$$\begin{array}{c} \\ 00 \\ 01 \\ 10 \\ 11 \end{array} \begin{array}{c} \begin{array}{cccc} 00 & 01 & 10 & 11 \end{array} \\ \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix} \end{array}.$$

- General controlled U-gate



$$|c\rangle|t\rangle \rightarrow |c\rangle U^c |t\rangle$$

Two qubit Gates

- CNOT gate : control bit

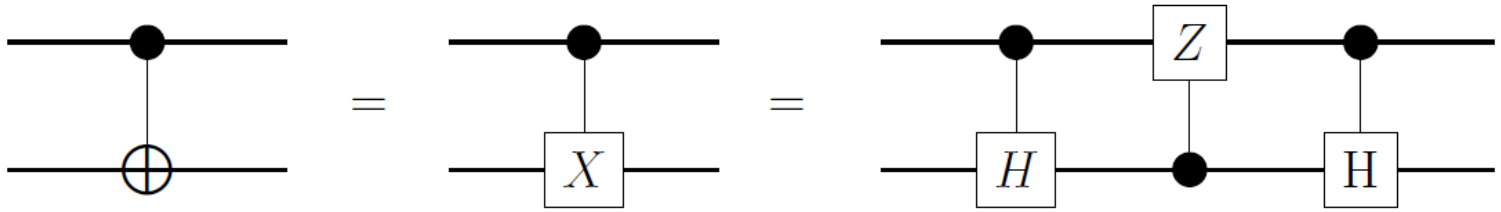
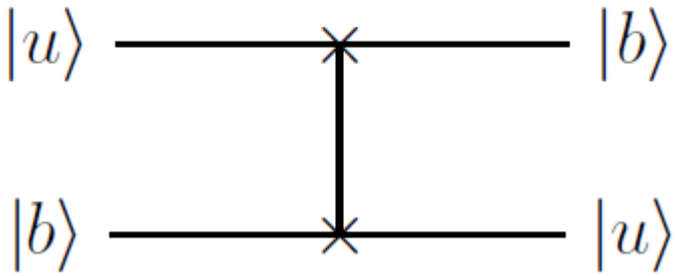


Fig. 17. Circuit identity.

- SWAP gate

$$|u\rangle|b\rangle \xmapsto{\text{SWAP}} |b\rangle|u\rangle$$



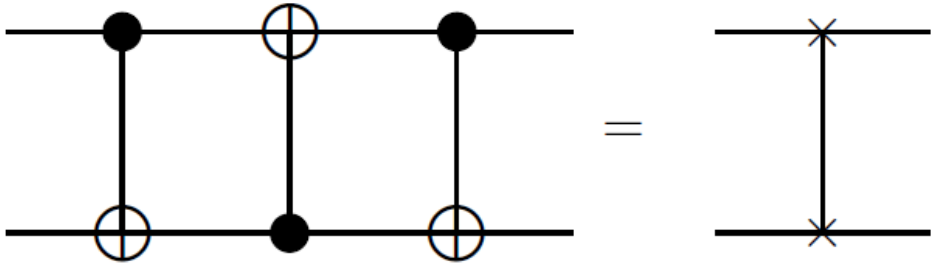
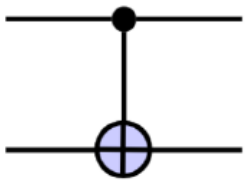
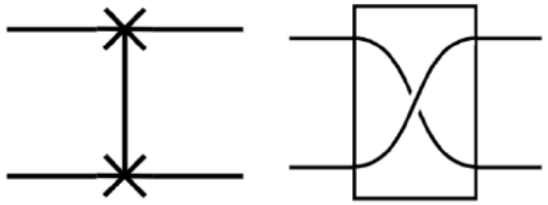
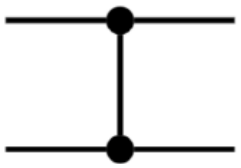
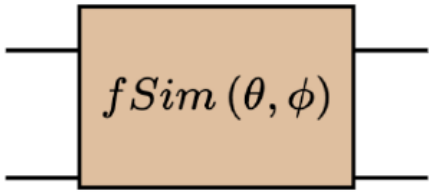
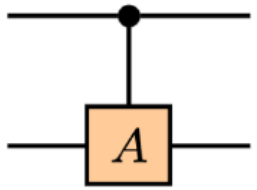
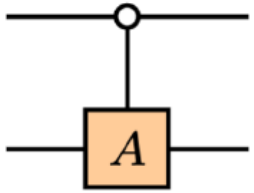
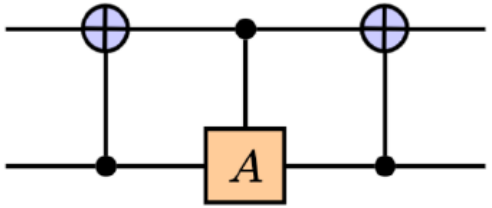
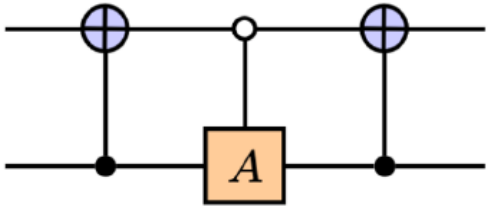
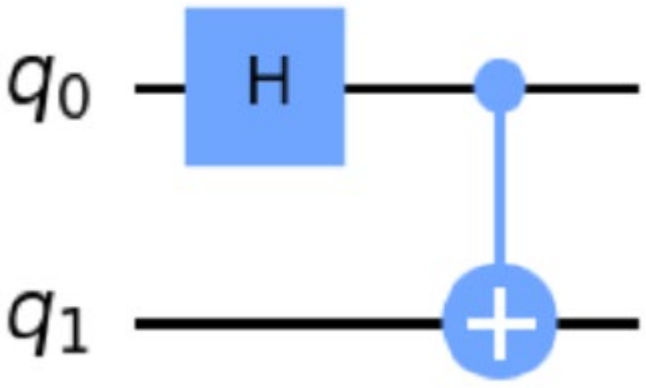


Fig. 21. $\text{CNOT}^3 = \text{SWAP}$.

Two qubit Gates

Name	Circuit	Matrix	Name	Circuit	Matrix
CNOT		$\begin{bmatrix} 1&0&0&0\\ 0&0&0&1\\ 0&0&1&0\\ 0&1&0&0 \end{bmatrix}$	SWAP		$\begin{bmatrix} 1&0&0&0\\ 0&0&1&0\\ 0&1&0&0\\ 0&0&0&1 \end{bmatrix}$
CZ		$\begin{bmatrix} 1&0&0&0\\ 0&1&0&0\\ 0&0&1&0\\ 0&0&0&-1 \end{bmatrix}$	fSim		$\begin{bmatrix} 1&0&0&0\\ 0&\cos(\theta)&-i\sin(\theta)&0\\ 0&-i\sin(\theta)&\cos(\theta)&0\\ 0&0&0&e^{i\phi} \end{bmatrix}$
a)		$\begin{bmatrix} 1&0&0&0\\ 0&a_{00}&0&a_{01}\\ 0&0&1&0\\ 0&a_{10}&0&a_{11} \end{bmatrix}$	b)		$\begin{bmatrix} a_{00}&0&a_{01}&0\\ 0&1&0&0\\ a_{10}&0&a_{11}&0\\ 0&0&0&1 \end{bmatrix}$
c)		$\begin{bmatrix} 1&0&0&0\\ 0&a_{00}&a_{01}&0\\ 0&a_{10}&a_{11}&0\\ 0&0&0&1 \end{bmatrix}$	d)		$\begin{bmatrix} a_{00}&0&0&a_{01}\\ 0&1&0&0\\ 0&0&1&0\\ a_{10}&0&0&a_{11} \end{bmatrix}$

Two qubit Gates



$$|0\rangle \otimes |0\rangle \equiv |00\rangle$$

$$|00\rangle \rightarrow (|0\rangle + |1\rangle)/\sqrt{2} \otimes |0\rangle \equiv (|00\rangle + |10\rangle)/\sqrt{2}$$

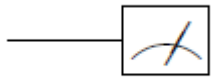
$$\left(|0\rangle\langle 0| \otimes I + |1\rangle\langle 1| \otimes X\right)\left(|0\rangle \otimes |0\rangle/\sqrt{2} + |1\rangle \otimes |0\rangle/\sqrt{2}\right)$$

$$= \left(|0\rangle \otimes |0\rangle/\sqrt{2} + |1\rangle \otimes |1\rangle/\sqrt{2}\right) \equiv (|00\rangle + |11\rangle)/\sqrt{2}$$

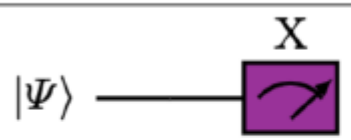
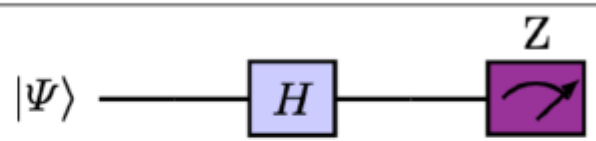
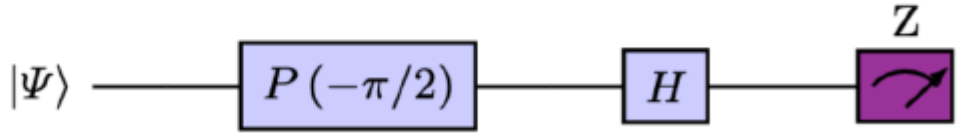
- The final state for the two qubits cannot be represented as a tensor product of single qubit states — it is an *entangled* state:

$$(|00\rangle + |11\rangle)/\sqrt{2} \neq (\alpha|0\rangle + \beta|1\rangle) \otimes (\gamma|0\rangle + \delta|1\rangle)$$

Measurements



Measurement

Measurement	Conversion to measurement in the Z-basis
	
	

Any measurement on single qu bit can be converted into Z-basis

The outcome of a measurement of $Z_{i_1} \otimes \cdots \otimes Z_{i_k}$ is a bitstring $b_{i_1} \ldots b_{i_k}$, with $b_i \in \{0, 1\}$. This bitstring is obtained with a probability given by Born's rule,

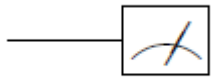
$$p(b_{i_1} \ldots b_{i_k}) = \langle \Psi | P_{b_{i_1} \ldots b_{i_k}} | \Psi \rangle, \tag{3.9}$$

$$P_{b_{i_1} \ldots b_{i_k}} = |b_{i_1}\rangle\langle b_{i_1}| \otimes \cdots \otimes |b_{i_k}\rangle\langle b_{i_k}|$$

For n-qubit system, k-qubits measurement Projects to the state

$$P_{b_{i_1} \ldots b_{i_k}} |\Psi\rangle / \sqrt{p(b_{i_1} \ldots b_{i_k})},$$

Measurements



Measurement

- We may measure all qubits to extract the probabilities of each basis bits strings
- Or, we may design the circuits so that the partial measurement of some qubits gives the desired information.

- Partial measurement in 2 qubit system

$$|q_2\rangle = \sum_{j,k} \alpha_{jk}|jk\rangle$$

$$P(|i\,\cdot\rangle) = P(|i\,0\rangle) + P(|i\,1\rangle) = |\langle i\,0|q_2\rangle|^2 + |\langle i\,1|q_2\rangle|^2$$
$$= |\alpha_{i0}|^2 + |\alpha_{i1}|^2 = \sum_j |\alpha_{ij}|^2 .$$

$$P(|\cdot\,k\rangle) = \sum_i |\alpha_{ik}|^2 .$$

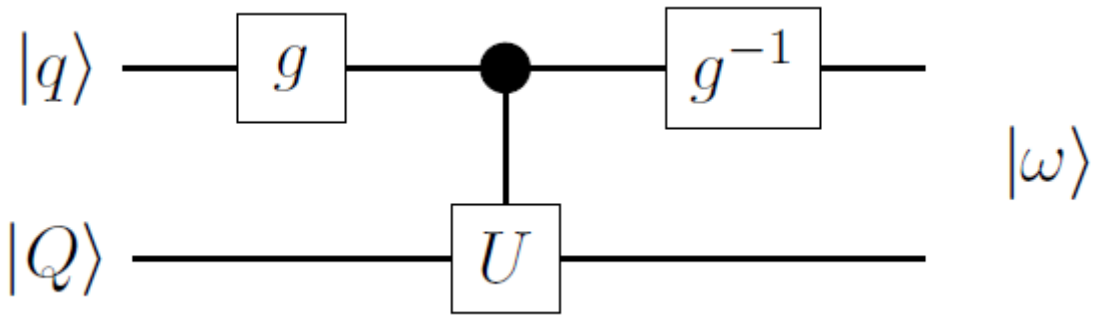
Measurements

- Partial measurement in multiple qubits
- Controlled-U gate

$$|i\rangle|Q\rangle \xmapsto{CU} |i\rangle U^i|Q\rangle$$

$$|q\rangle|Q\rangle = \sum_i \alpha_i|i\rangle|Q\rangle \xmapsto{CU} \sum_i \alpha_i|i\rangle U^i|Q\rangle \,.$$

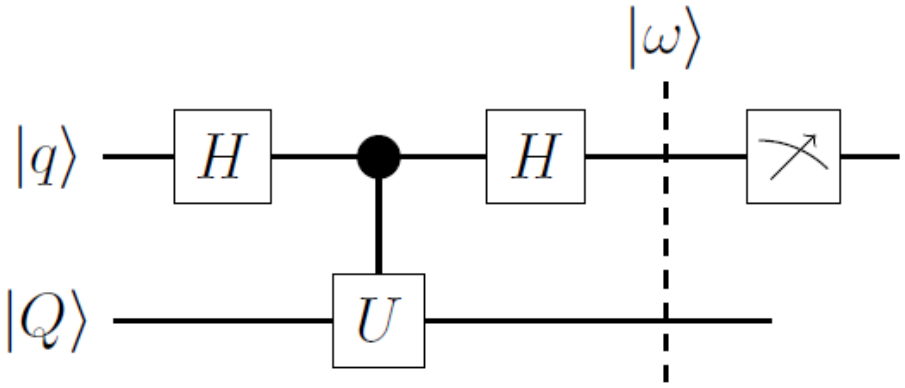
Measurements



$$|q\,Q\rangle \xmapsto{g} \sum_{i,j} g_{ij}\alpha_i|i\,Q\rangle \xmapsto{CU} \sum_{i,j} g_{ij}\alpha_j U^i|i\,Q\rangle \xmapsto{g^{-1}} \sum_{i,j,k} g^*_{ki}g_{kj}\alpha_j U^k|i\,Q\rangle\,.$$

$$\begin{aligned}|\omega\rangle = &\left[\alpha_0(g^*_{00}g_{00} + g^*_{10}g_{10}U) + \alpha_1(g^*_{00}g_{01} + g^*_{10}g_{11}U)\right]|0\,Q\rangle \\ &+ \left[\alpha_0(g^*_{01}g_{00} + g^*_{11}g_{10}U) + \alpha_1(g^*_{01}g_{01} + g^*_{11}g_{11}U)\right]|1\,Q\rangle\,.\end{aligned}$$

Measurements



If control qubit is prepare in |0>

$$|\omega\rangle = \frac{1}{2}(1+U)|0\,Q\rangle + \frac{1}{2}(1-U)|1\,Q\rangle$$

$$(1 \pm U) \text{ we really mean } (1 \otimes 1 \pm 1 \otimes U)$$

quantum phase estimation algorithm.

If |Q> is an eigenstate of U,
We can measure eigenvalue of |Q>

$$U|Q\rangle = e^{i\theta}|Q\rangle$$

$$|\omega\rangle = \frac{1}{2}(1+e^{i\theta})|0\,Q\rangle + \frac{1}{2}(1-e^{i\theta})|1\,Q\rangle$$

$$P(|0\,\cdot\rangle) = \frac{1}{4}(1+e^{i\theta})(1+e^{-i\theta}) = \cos^2(\theta/2)$$

$$P(|1\,\cdot\rangle) = \frac{1}{4}(1-e^{i\theta})(1-e^{-i\theta}) = \sin^2(\theta/2)$$

Deutsch's Algorithm

- The first algorithm which shows quantum supremacy.
- For given single bit Boolean function, determine whether it is constant or balanced.

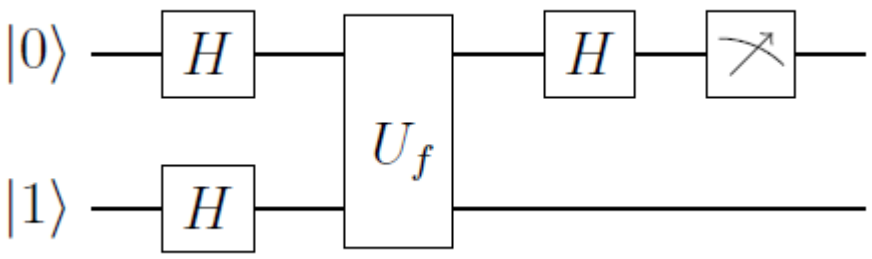
$$f : \{0, 1\} \rightarrow \{0, 1\}$$

$f(0) = f(1)$ constant

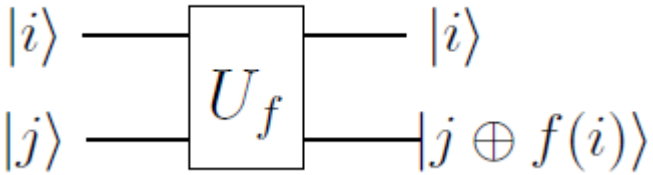
$f(0) \neq f(1)$ balenced

- Classical circuits require two measurements (or queries) to answer the question.
- Quantum circuits require one query.

Deutsch's Algorithm



XOR oracle



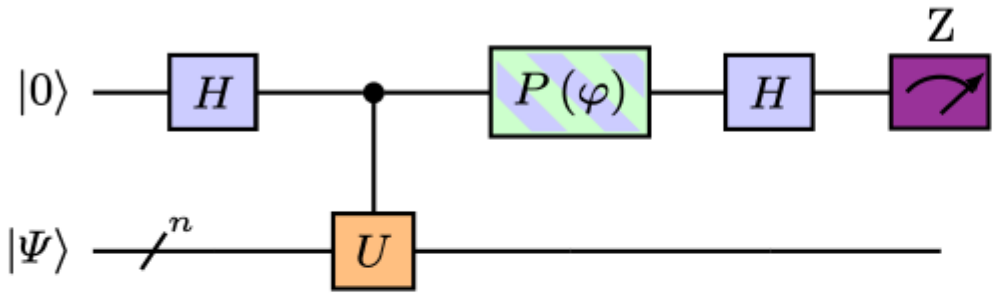
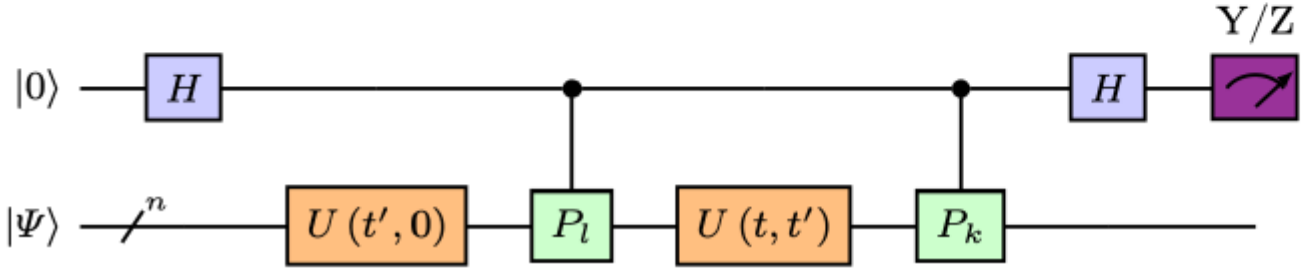
$$|0\,1\rangle \xmapsto{H\otimes H} |+\,-\rangle$$

$$U_{fc}|+\,-\rangle \xmapsto{H\otimes 1} \pm(H\otimes 1)|+\,-\rangle = \pm|0-\rangle$$

$$U_{fb}|+\,-\rangle \xmapsto{H\otimes 1} \pm(H\otimes 1)|-\,-\rangle = \pm|1-\rangle$$

Thus,
if we measure |0> → f is constant
If we measure |1> → f is balanced

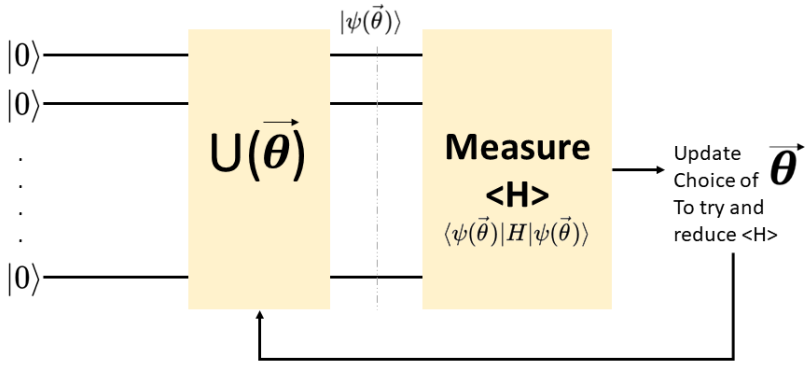
Circuit measurement

Circuit $\mathcal{C}$	Calculated overlap
	$\langle\Psi\|U\|\Psi\rangle = \langle Z_0\rangle_{\mathcal{C}(\varphi=0)} + i\langle Z_0\rangle_{\mathcal{C}(\varphi=\pi/2)}$
	$\langle\Psi\|U^\dagger(t,0)P_kU(t,t')P_lU(t',0)\|\Psi\rangle$ $= \langle Z_0\rangle_{\mathcal{C}} + i\langle Y_0\rangle_{\mathcal{C}}$

We can measure expectation value of unitary operator of entangled qubits |Psi>

Here <Z0> means average value of 0-th qubit measurement in Z-basis.

Variational Quantum Eigensolver



1. Design Circuits so that the measurement corresponds to t he expectation value of Hamiltonian
2. Design Circuits so that the Unitary operator parameter co rresponds to give variational wave function
3. Measure energy expectation value
4. update parameter to minimize the Hamiltonian expectatio n value
5. we have variational optimal wave function.

Simple VQE example

FEATURED IN PHYSICS | EDITORS' SUGGESTION

Cloud Quantum Computing of an Atomic Nucleus

E. F. Dumitrescu[1], A. J. McCaskey[2], G. Hagen[3,4], G. R. Jansen[5,3], T. D. Morris[4,3], T. Papenbrock[4,3,*], R. C. Pooser[1,4], D. J. Dean[3], and P. Lougovski[1,†]

Show more

Phys. Rev. Lett. **120**, 210501 – **Published 23 May, 2018**

DOI: https://doi.org/10.1103/PhysRevLett.120.210501

$$H_N = \sum_{n,n'=0}^{N-1} \langle n'|(T+V)|n\rangle a^\dagger_{n'} a_n.$$

- Deuteron eigenvalue in HO basis
  - Two-body
  - Consider only S-wave
  - HO basis
  - Pionless EFT

$$\langle n'|T|n\rangle = \frac{\hbar\omega}{2}\left[(2n+3/2)\delta^{n'}_n - \sqrt{n(n+1/2)}\delta^{n'+1}_n - \sqrt{(n+1)(n+3/2)}\delta^{n'-1}_n\right]$$

$$\langle n'|V|n\rangle = V_0 \delta^0_n \delta^{n'}_n$$

$$V_0 = -5.68658111\text{MeV},$$

$$\hbar\omega = 7\text{MeV}.$$

Simple VQE example

N=2 2-qubit system $\{|\downarrow\uparrow\rangle, |\uparrow\downarrow\rangle\}$

- Encoding:
  - Jordan-Wigner transformation :

  creation/annihilation operators → Pauli operator sequences

$$a_n^\dagger \to \frac{1}{2}\left[\prod_{j=0}^{n-1} -Z_j\right](X_n - iY_n)$$

$$a_n \to \frac{1}{2}\left[\prod_{j=0}^{n-1} -Z_j\right](X_n + iY_n).$$

$$H_2 = 5.906709I + 0.218291Z_0 - 6.125Z_1 - 2.143304\,(X_0X_1 + Y_0Y_1)$$

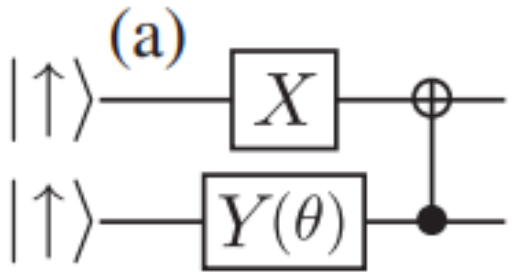
- However, Jordan-Wigner transformation or one-hot-encoding is not resource efficient.

Simple VQE example

- Variational Ansatz (Unitary coupled cluster Ansatz)

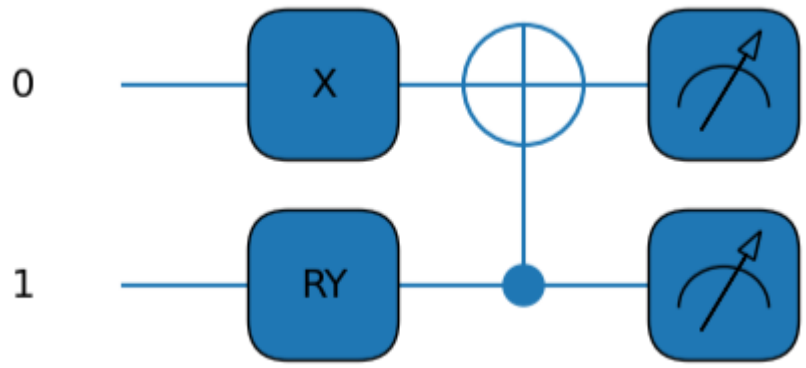
$$|\psi\rangle = \sum_{n=0}^{N-1} c_n |n\rangle = \sum_{x=(N-bit-string)} c_n(x)|x\rangle \qquad \{|\downarrow\uparrow\rangle, |\uparrow\downarrow\rangle\}$$

$$U(\theta) \equiv e^{\theta(a_0^\dagger a_1 - a_1^\dagger a_0)} = e^{i(\theta/2)(X_0 Y_1 - X_1 Y_0)}$$

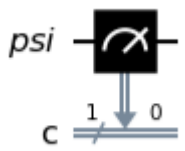


Simple VQE example

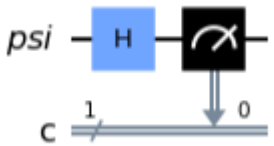
VQE circuits (Ansatz + Hamiltonian)



$$H_2 = 5.906709 I + 0.218291 Z_0 - 6.125 Z_1 - 2.143304 \left(X_0 X_1 + Y_0 Y_1\right)$$



<Z> measurement



<X> measurement
= <HZH>

<Y> measurement
= <S X S^dagger >

Simple VQE example

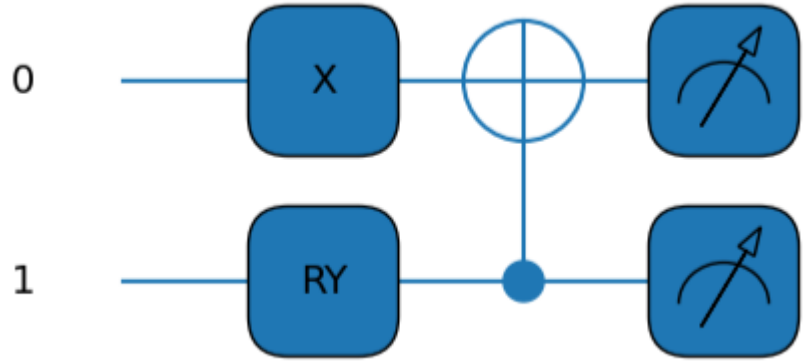
VQE circuits (Ansatz + Hamiltonian)

Python Qiskit + pennylane library

N=3 Hamiltonian circuits gives -2.0451 MeV

```
5.906709 * I(0) + 0.218291 * Z(0) + -6.125 * Z(1) + -2.143304 * (X(0) @ X(1)) + -2.143304 * (Y(0) @ Y(1))
(<Figure size 500x300 with 1 Axes>, <Axes: >)
Iteration = 0,  Energy = 7.89426366 MeV,  Convergence parameter = 0.52891694 MeV
Iteration = 20,  Energy = -0.66950925 MeV,  Convergence parameter = 0.17005461 MeV
Iteration = 40,  Energy = -1.70101182 MeV,  Convergence parameter = 0.00828148 MeV
Iteration = 60,  Energy = -1.74716419 MeV,  Convergence parameter = 0.00034480 MeV
Iteration = 80,  Energy = -1.74907865 MeV,  Convergence parameter = 0.00001426 MeV

Final value of the energy = -1.74915572 MeV
Number of iterations =  97
```



Simple VQE code example

- In fact, 2x2 matrix eigenvalue problem can be solved with only 1 qubit

$$H = \begin{pmatrix} 1 & 0.5 \\ 0.5 & 2 \end{pmatrix} \qquad H = 1.5I + 0.5X - 0.5Z \qquad |\psi(\theta)\rangle = \cos(\theta/2)|0\rangle + \sin(\theta/2)|1\rangle = \begin{pmatrix} \cos(\theta/2) \\ \sin(\theta/2) \end{pmatrix}$$

```python
import pennylane as qml
from pennylane import numpy as np

dev = qml.device("default.qubit", wires=1)

# Define the Hamiltonian: 1.5*I + 0.5*X - 0.5*Z
coeffs = [1.5, 0.5, -0.5]
obs = [qml.Identity(0), qml.PauliX(0), qml.PauliZ(0)]
H = qml.Hamiltonian(coeffs, obs)

@qml.qnode(dev)
def circuit(theta):
    qml.RY(theta, wires=0)  # The Ansatz
    return qml.expval(H)

# Optimization loop
opt = qml.GradientDescentOptimizer(stepsize=0.4)
theta = np.array(0.0, requires_grad=True)

for i in range(20):
    theta = opt.step(circuit, theta)

print(f"Final Energy: {circuit(theta):.4f}")

qml.draw_mpl(circuit,decimals=2)(theta=2.35)
plt.show()
```

```
Final Energy: 0.7929
```

Untouched subjects

- How to simplify the Hamiltonian to reduce required qubits?
- How to encode the physical basis into qubits ? How to convert Hamiltonian to Pauli matrices?
- How to make a good ansatz of wave function in VQE?
- How to construct efficient quantum circuits ?
- How to remove noises in the quantum measurements?
- Other quantum algorithms for nuclear physics?
- Quantum simulators
- Time evolution, Trotterization
- And Many more